ECE551: Digital System Design and Synthesis

Fall 2026

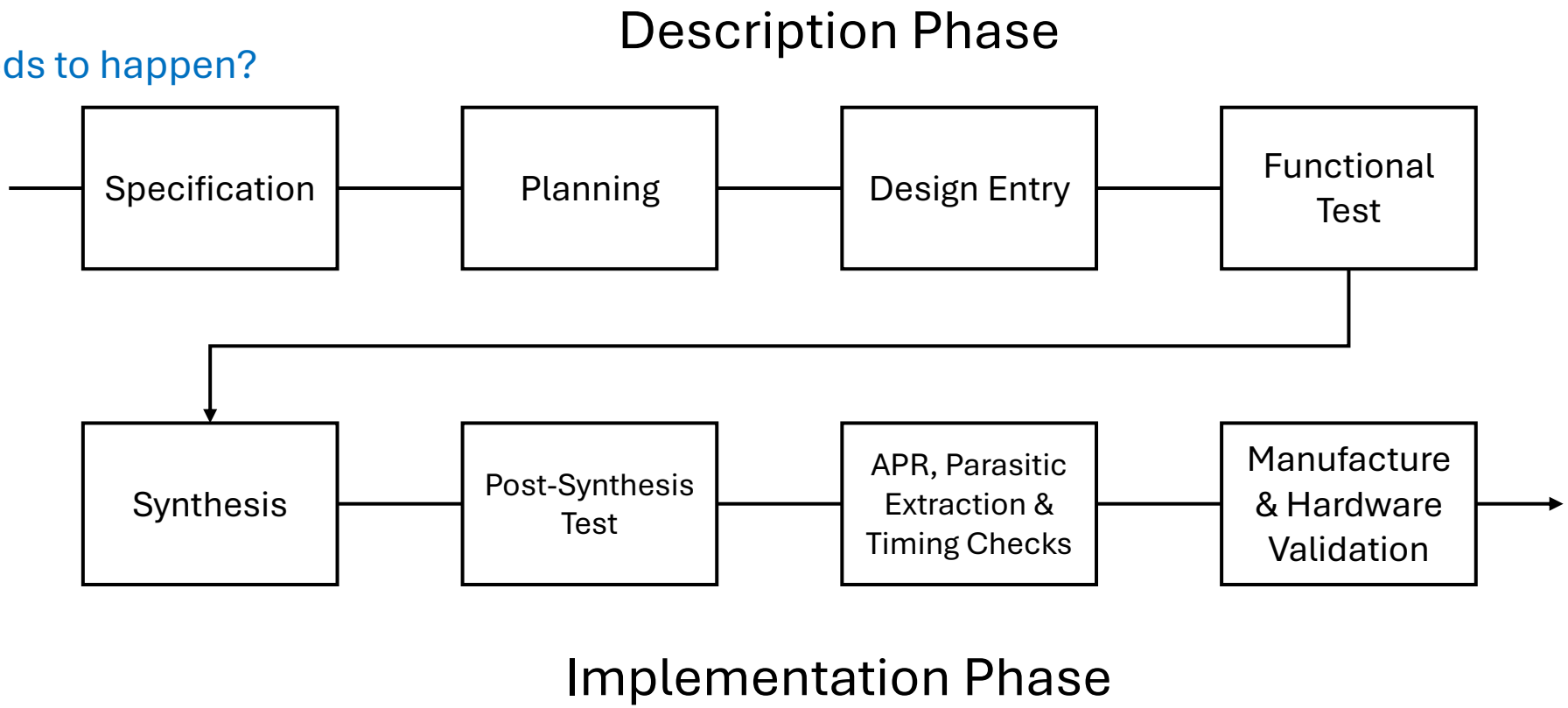
Digital Design Process



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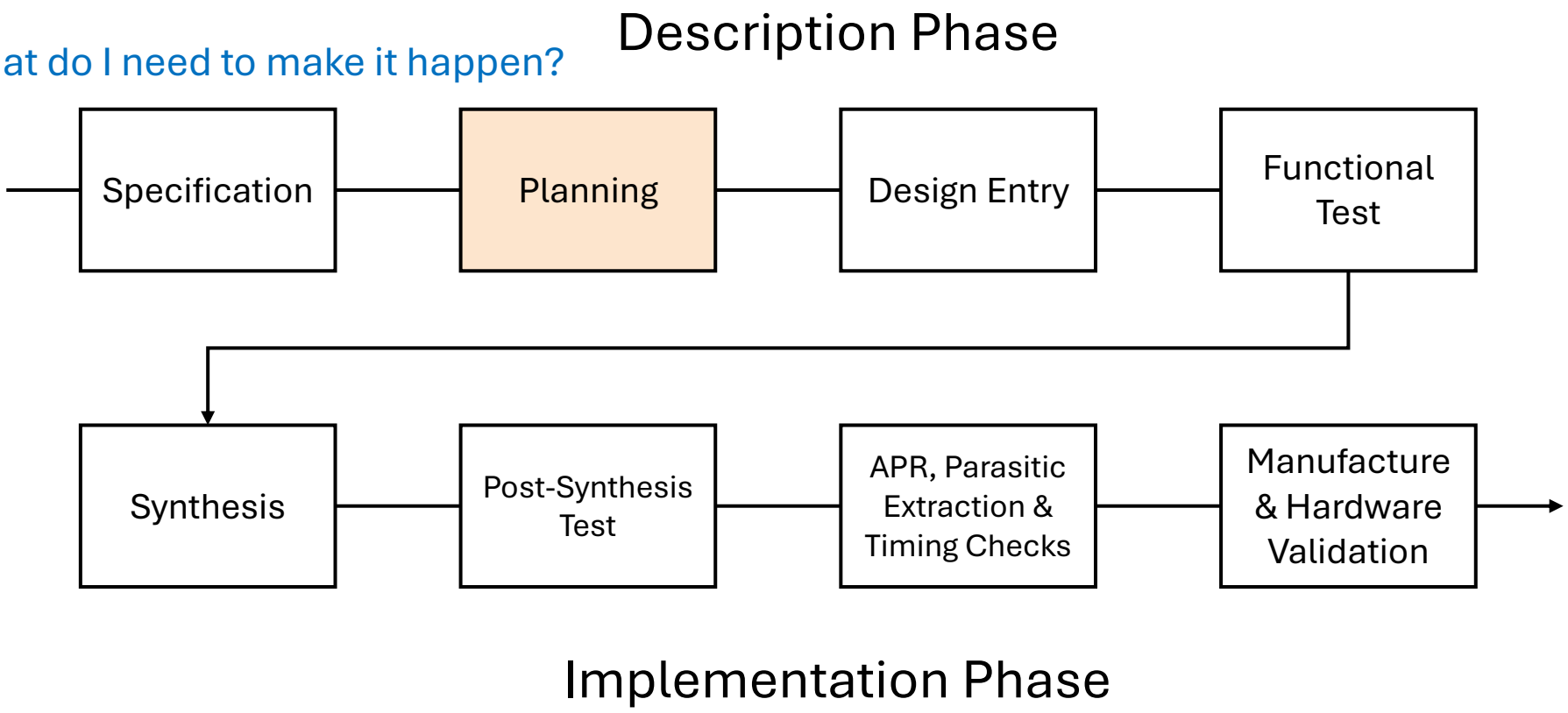
What needs to happen?



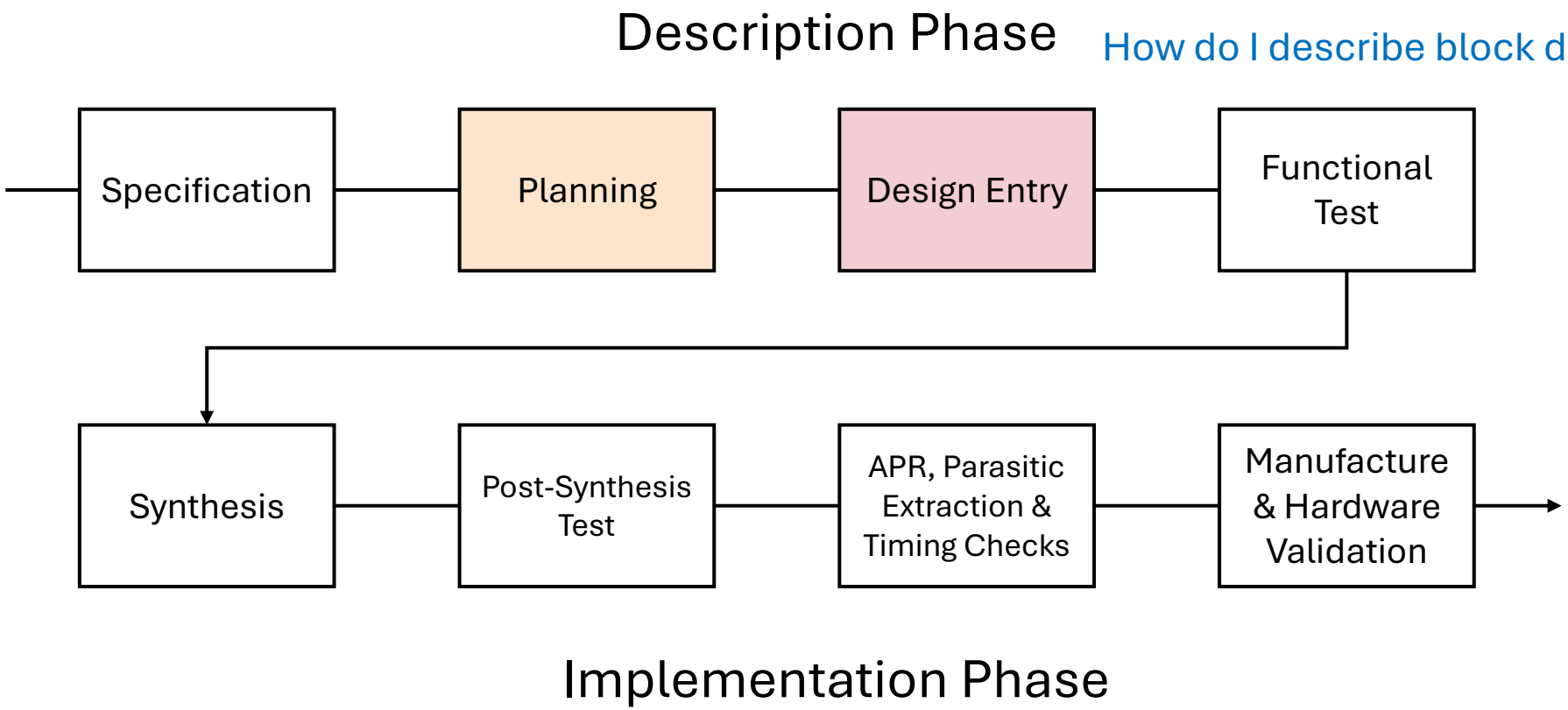
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What do I need to make it happen?



Digital Design Process



What is a Hardware Description Language (HDL)?

- Describes the structure and behavior of hardware circuits
- Allows for modeling & simulation (with timing) of digital designs
- Can be synthesized into hardware (netlist) by synthesis tools (Synopsys, Cadence, FPGA compilers)
- Two major standards in industry & academia
  - Verilog/SystemVerilog (Flexible, loose, more common in industry)
  - VHDL (Strongly typed, more common in defense and automotive)

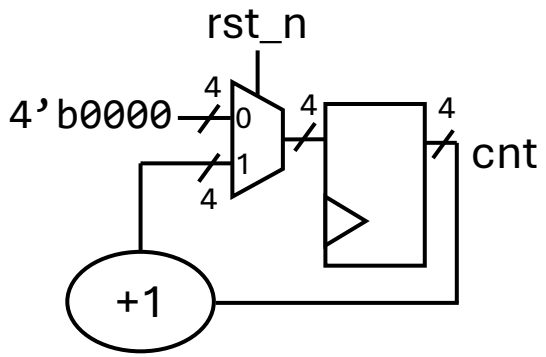
What is a Hardware Description Language (HDL)?

```
module counter(clk,rst_n,cnt);
  input logic clk,rst_n;
  output logic [3:0] cnt;

  always_ff @(posedge clk) begin
    if (!rst_n)
       cnt <= 4'b0000;
    else
       cnt <= cnt+1;
  end
endmodule
```

- Looks like a programming language, but it is NOT
- Remember that this is used to **describe hardware**
  - Think "what hardware does this map to?" when reading Verilog
  - Code should clearly describe the structure & behavior of hardware
  - We will develop good coding practice to make mapping of Verilog ←→ hardware as straight forward as possible

- This example:
  - Block where something happens at posedge of clk → Flipflop
  - If-else → multiplexer

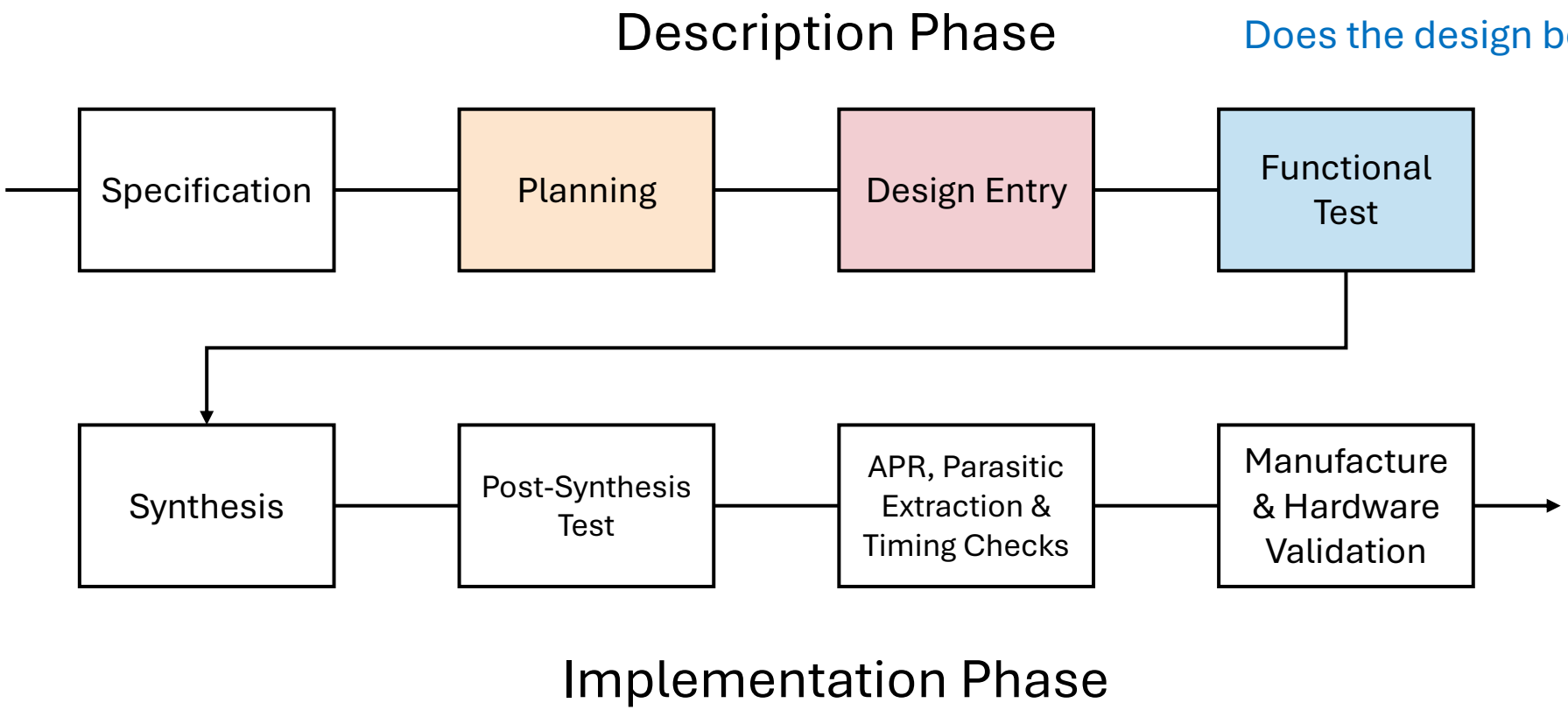


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Why Use an HDL?

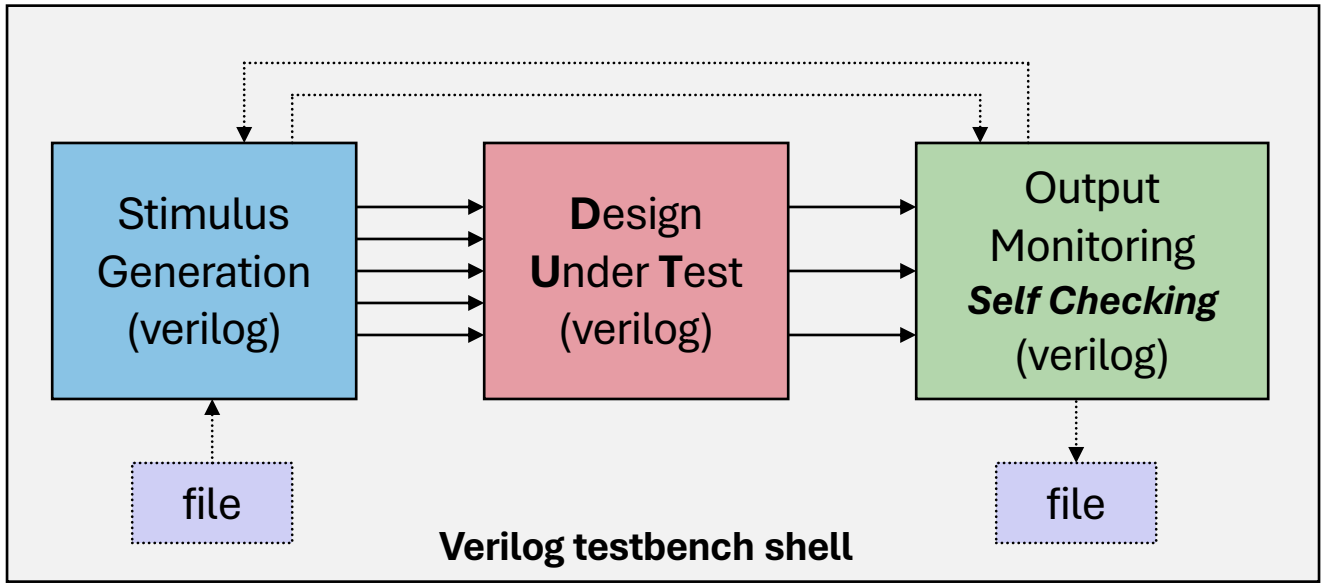
- Enables larger designs – higher level of abstraction
  - Wide datapaths (16, 32, or 64 bits wide) can be abstracted to a single vector
  - Register Transfer Level (RTL) description
  - Synthesis tool does the bulk of the tedious repetitive work
- Portable Design
  - Behavioral or dataflow Verilog can be synthesized to a new process library with little effort (i.e., move from 45nm to 22nm process)
  - Verilog in ASCII text can be used by synthesis tools from multiple vendors
- Explore larger solution space
  - Synthesis options and coding styles can help optimize and examine tradeoffs (power, area, speed)
- Better Validated Designs

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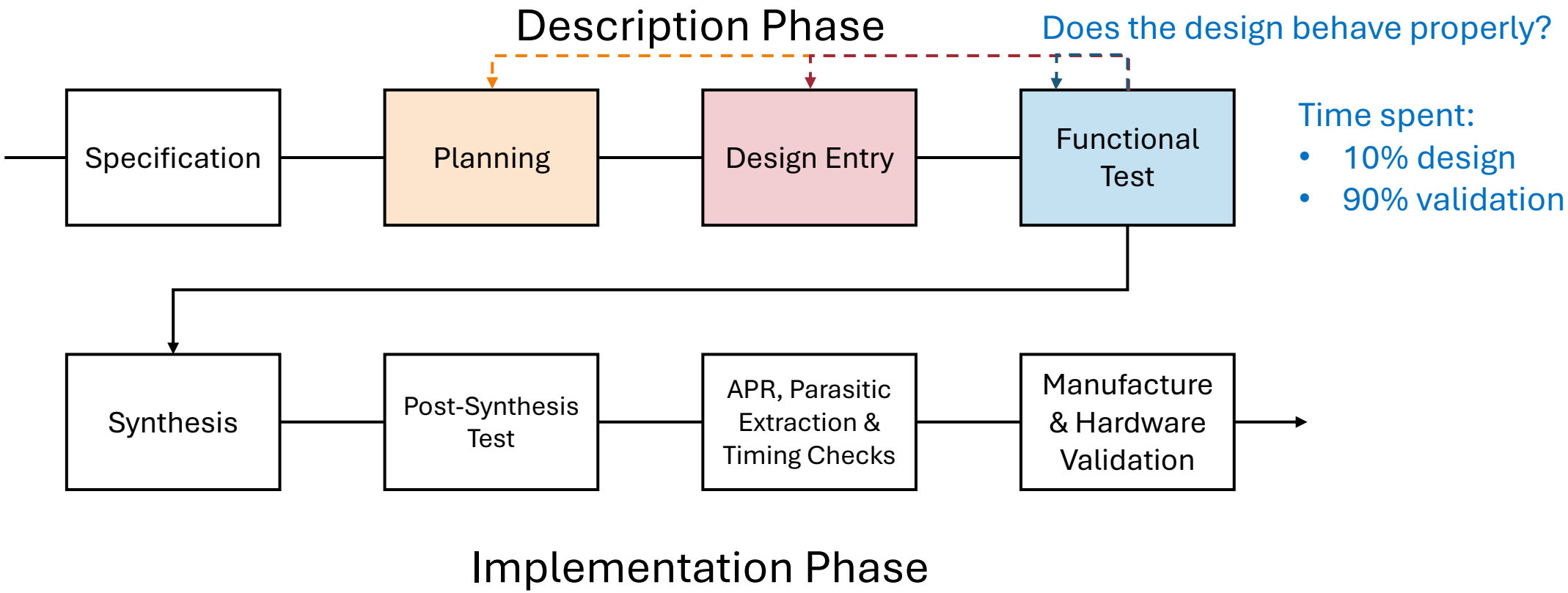
Simulating / Validating HDL

- Testbenches provide input stimulus to your design
- Monitor outputs from your design

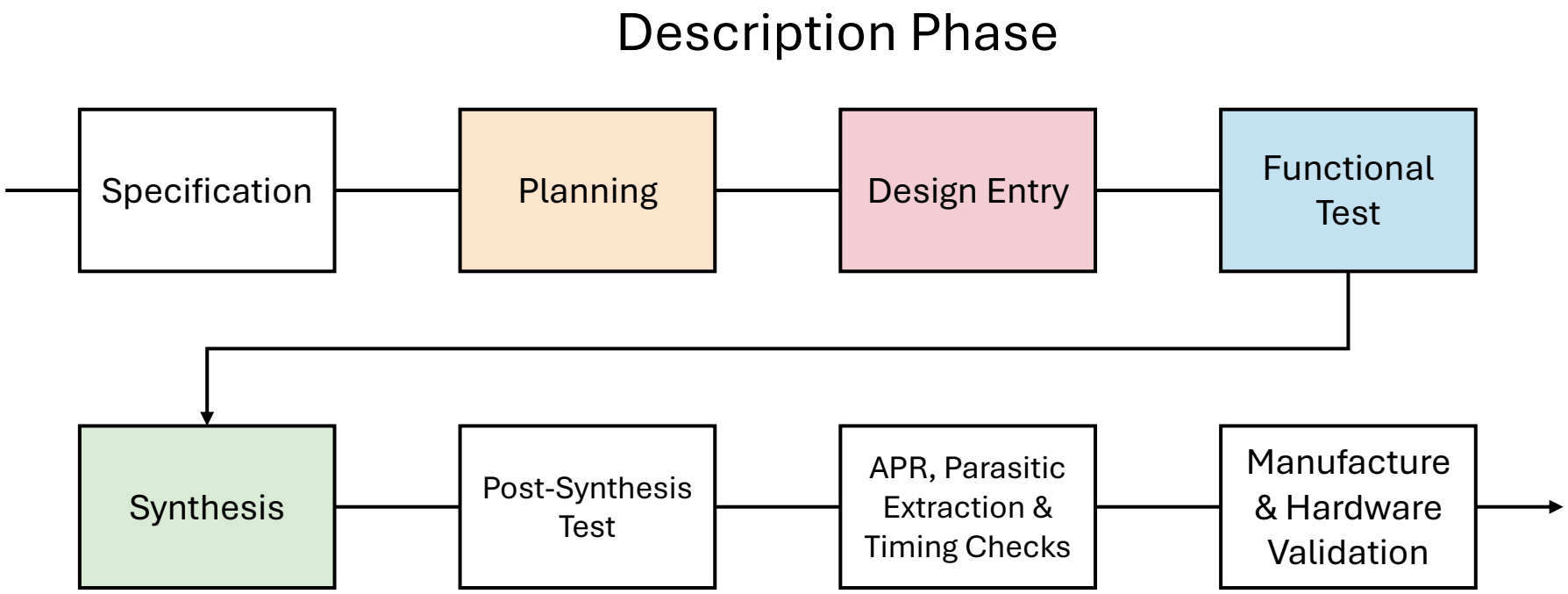


- Testbenches are written in Verilog as well… but does not describe hardware
  - Testbench Verilog describes sequence of events → more similar to a program

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Digital Design Process



What is Synthesis?

- Takes a description of what a circuit DOES and create hardware to DO it

```
module circuitF(a,b,c,d,e,f);

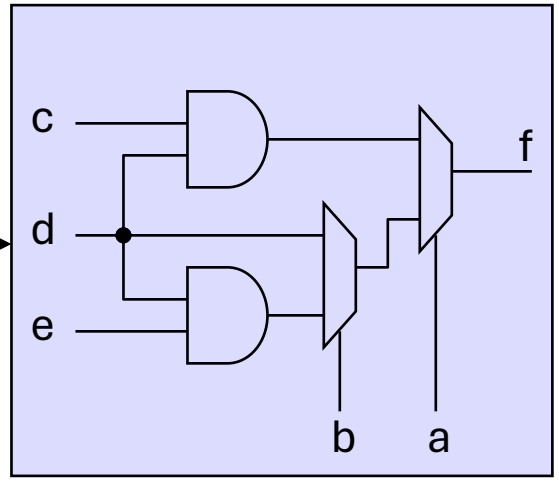
  input a,b,c,d,e;
  output f;

  always_comb begin
    if (a)      f = c & d;
    else if (b) f = d;
    else        f = d & e;
  end
endmodule
```

Synthesis Tool (Synopsys)

Timing Constraints

*Output is actually a text netlist, not a GUI schematic form.



Estimated power, performance, area

What is a Netlist?

- An ASCII text representation of a schematic
- Many standards exist:
  - SPICE Netlist
  - EDIF (Electronic Data Interchange Format)
  - Structural Verilog Netlist (this is what we will use)

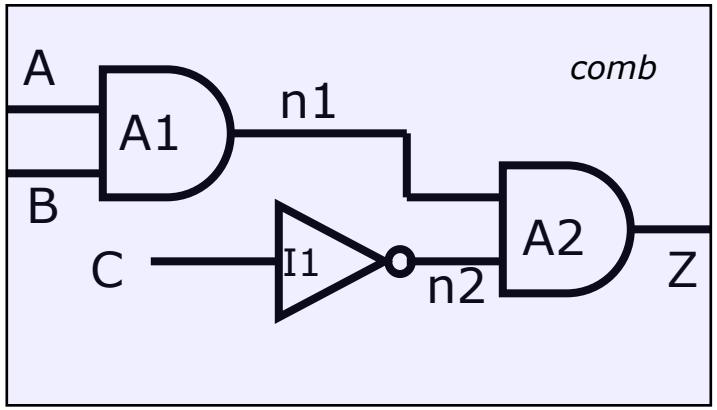
```
module comb(Z,A,B,C);

  input A,B,C;
  output Z;
  wire n1, n2;

  and02d1 A1(n1,A,B);
  inv01d1  I1(n2,C);
  and02d2 A2(Z,n1,n2);

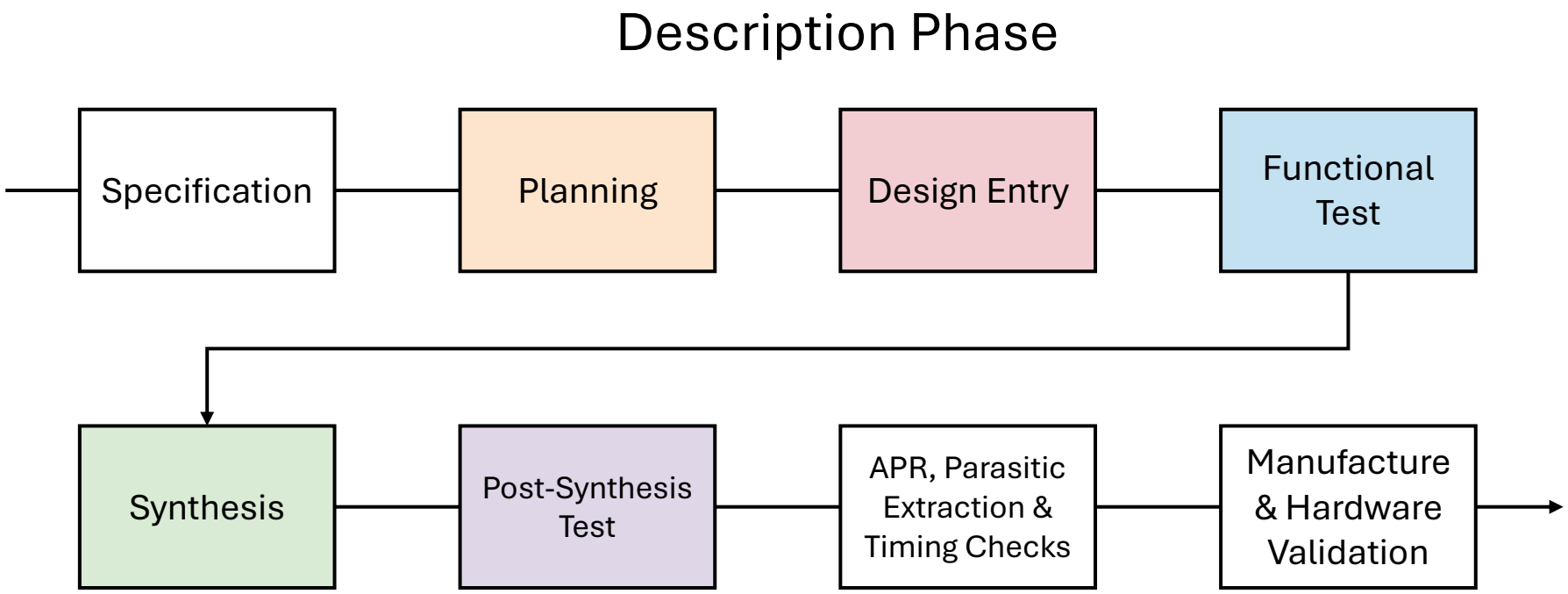
endmodule
```

=

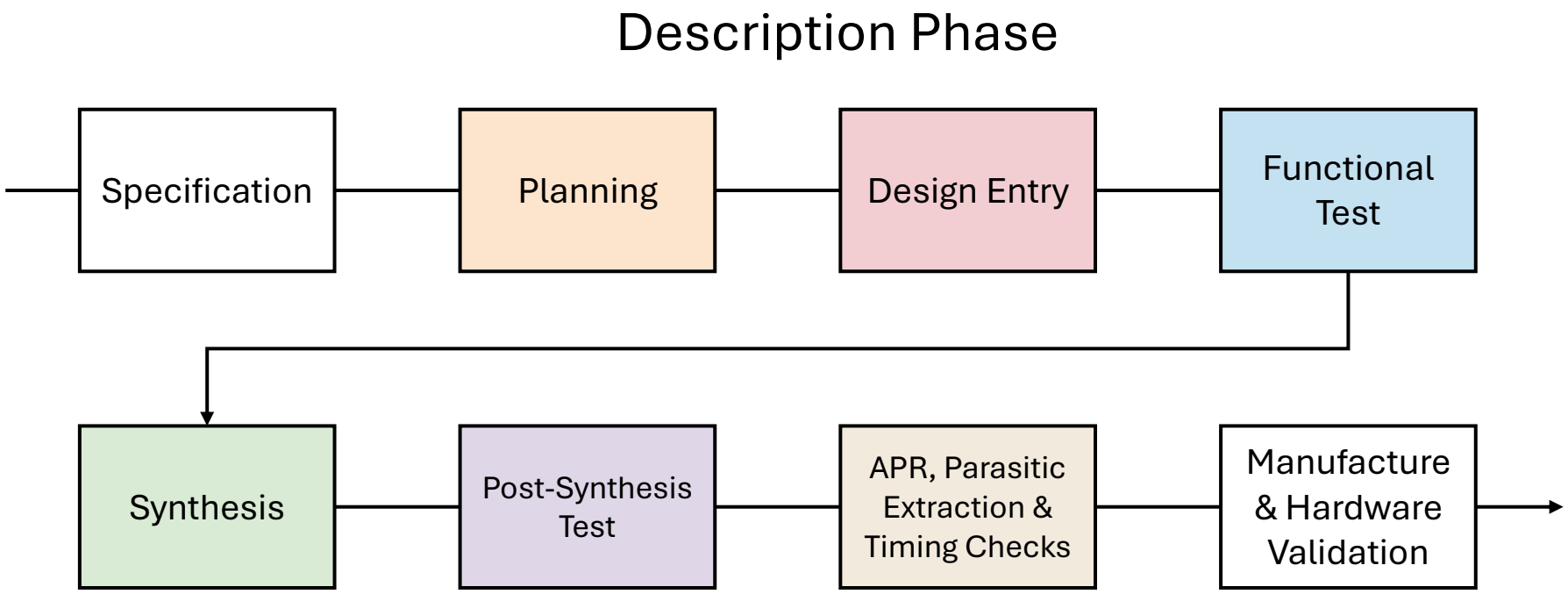


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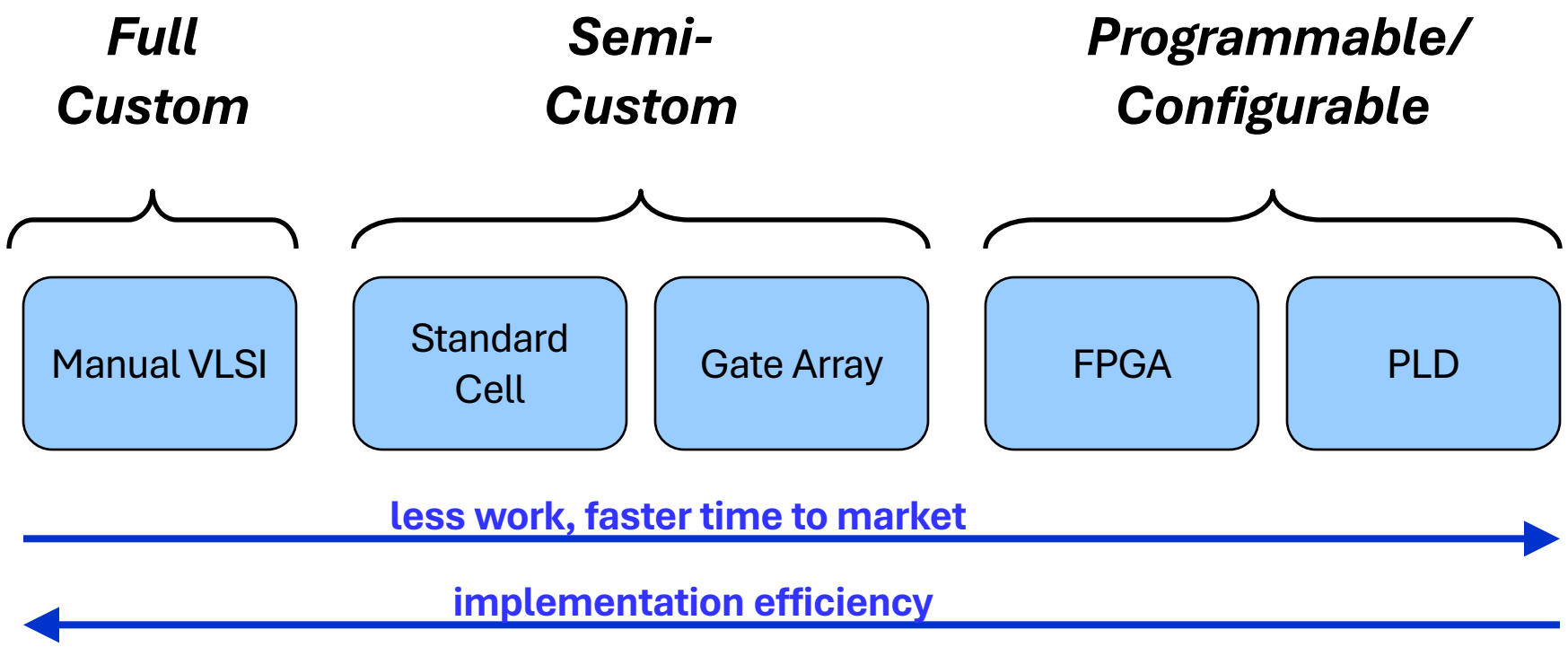


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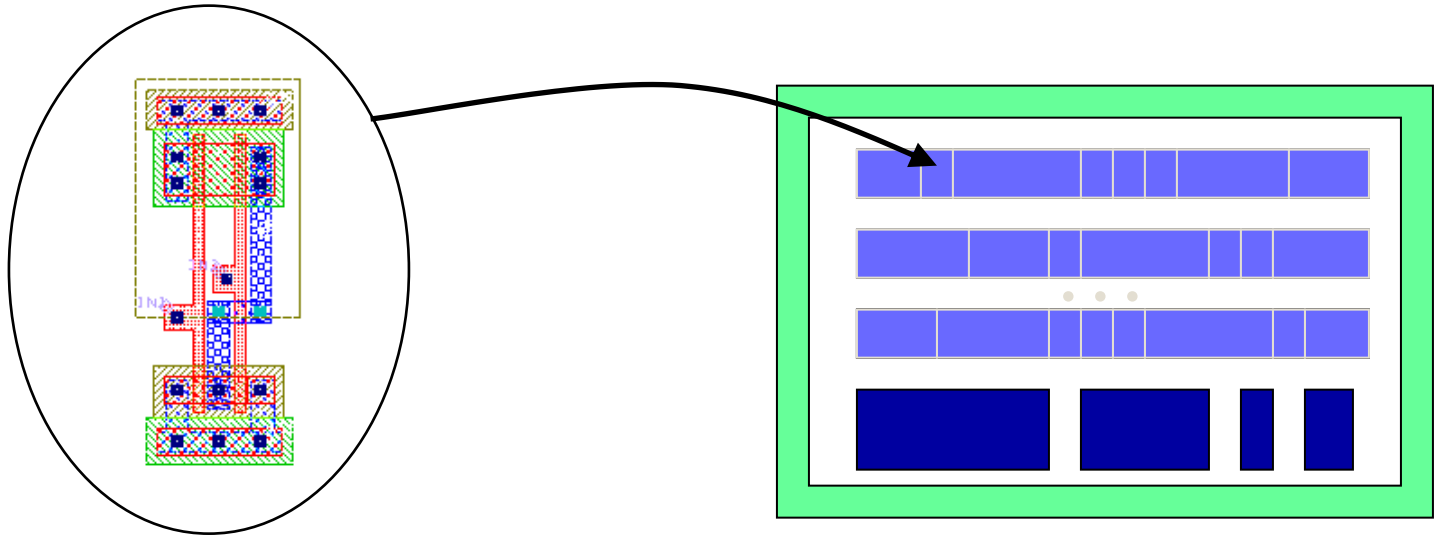
Hardware Implementations

- HDLs can be compiled to semi-custom and programmable hardware implementations



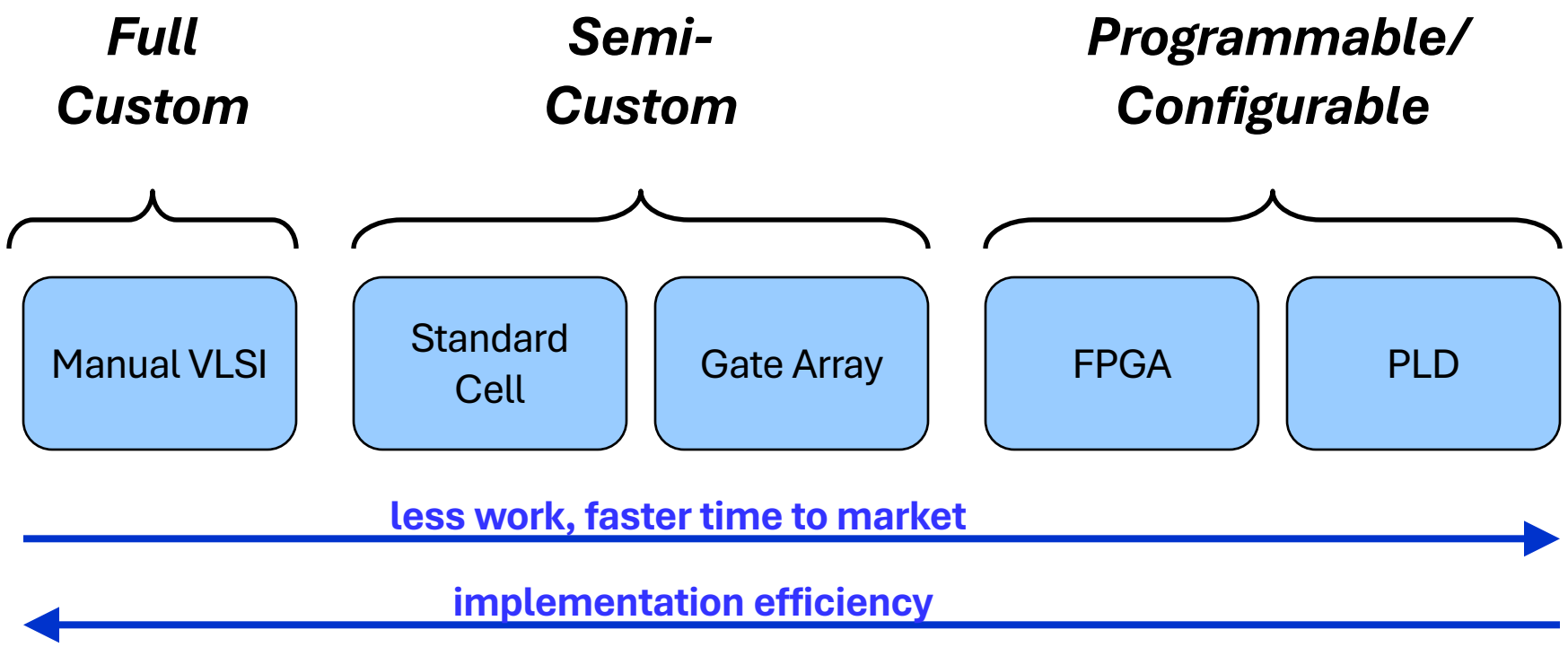
Standard Cells

- Library of common gates and structures (cells)
- Decompose hardware in terms of these cells
- Arrange the cells on the chip
- Connect them using metal wiring



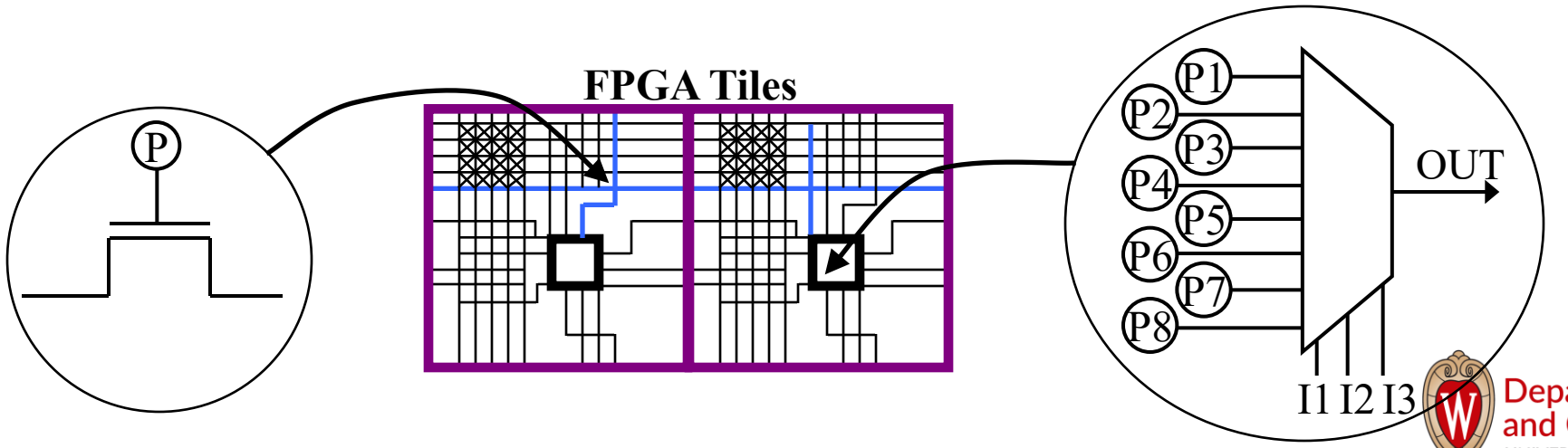
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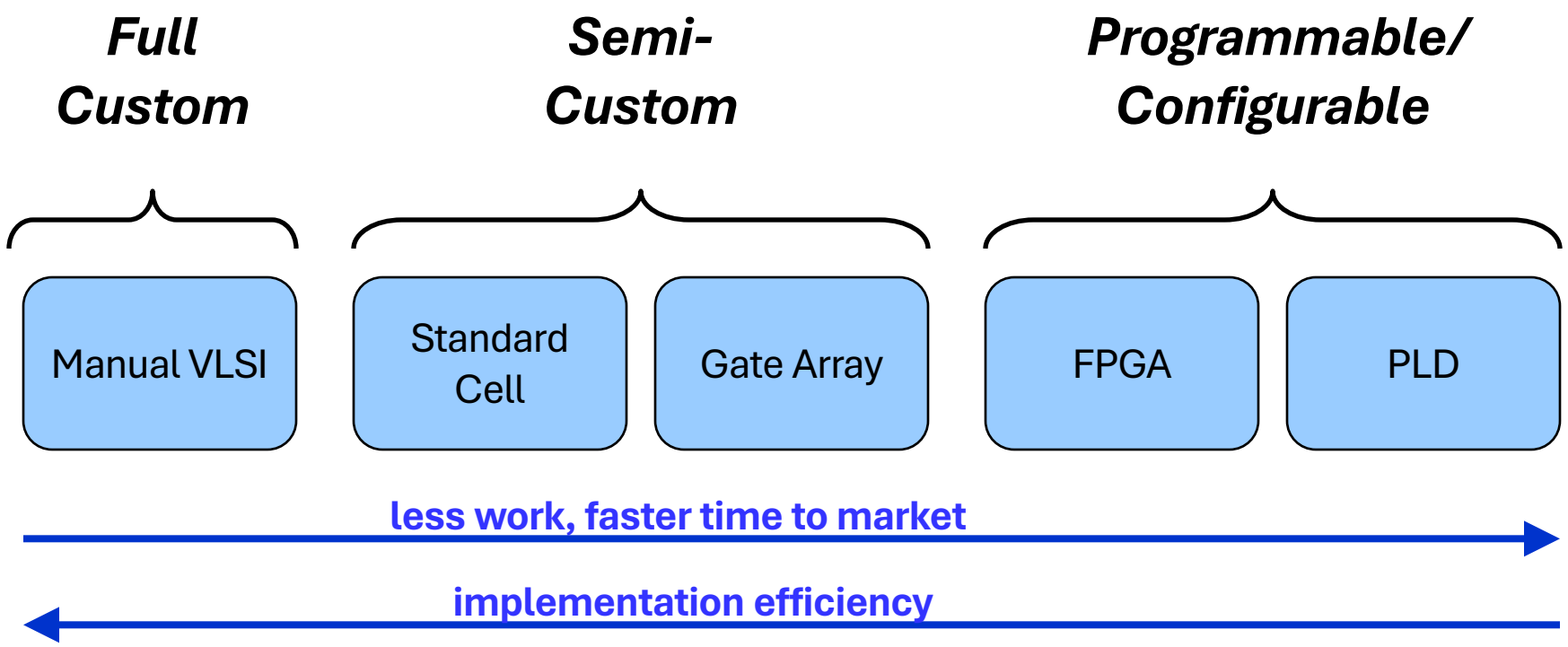
Field Programmable Gate Arrays (FPGAs)

- “Programmable” hardware
- Use small memories as truth tables of functions
- Decompose circuit into these blocks
- Connect using programmable routing
- SRAM bits control functionality



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